

# PLS-SEM Reporting Checklist

Every required output for a defensible PLS-SEM manuscript, mapped to Hair et al. thresholds.

## STEP 1 REFLECTIVE MEASUREMENT MODEL

| METRIC                                                  | THRESHOLD                                       | IF IT FAILS                                          |
|---------------------------------------------------------|-------------------------------------------------|------------------------------------------------------|
| <input type="checkbox"/> Indicator loadings             | $\geq 0.708$                                    | Consider dropping items with loadings $< 0.40$       |
| <input type="checkbox"/> Cronbach's alpha               | $0.70 - 0.95$                                   | Below $0.60$ = inadequate; above $0.95$ = redundancy |
| <input type="checkbox"/> Composite reliability $\rho_C$ | $0.70 - 0.95$                                   | Report $\rho_A$ as well (Dijkstra-Henseler)          |
| <input type="checkbox"/> AVE (convergent validity)      | $\geq 0.50$                                     | Drop weakest loader; check theoretical fit           |
| <input type="checkbox"/> HTMT (discriminant validity)   | $< 0.85$ strict / $0.90$ lib.                   | Report bootstrap CI; must exclude $1.0$              |
| <input type="checkbox"/> Fornell-Larcker criterion      | $\sqrt{\text{AVE}} > \text{inter-construct } r$ | HTMT is preferred; report both if space              |
| <input type="checkbox"/> Cross-loadings                 | Within-construct $>$ cross-load                 | Locate the offending indicator pair                  |

## STEP 2 FORMATIVE MEASUREMENT MODEL (if applicable)

| METRIC                                                    | THRESHOLD                  | IF IT FAILS                                           |
|-----------------------------------------------------------|----------------------------|-------------------------------------------------------|
| <input type="checkbox"/> Convergent validity (redundancy) | Path $\geq 0.70$           | Requires a global single-item proxy                   |
| <input type="checkbox"/> Indicator collinearity (VIF)     | $< 3$ (tolerant $\leq 5$ ) | Merge or drop redundant indicators                    |
| <input type="checkbox"/> Indicator weights (bootstrapped) | Significant at $p < 0.05$  | Retain non-sig. weights if loading $\geq 0.50$        |
| <input type="checkbox"/> Indicator loadings (retention)   | $\geq 0.50$ to keep        | Below $0.50$ with non-sig weight $\rightarrow$ remove |

*Skip this section if all constructs are reflective.*

## STEP 3 STRUCTURAL MODEL

| METRIC                                                       | THRESHOLD                       | IF IT FAILS                           |
|--------------------------------------------------------------|---------------------------------|---------------------------------------|
| <input type="checkbox"/> Path coefficient collinearity (VIF) | $< 3$ (tolerant $\leq 5$ )      | Merge overlapping predictors          |
| <input type="checkbox"/> Path coefficients + significance    | p-values via 5,000 bootstraps   | Report BCa 95% CI, not just p         |
| <input type="checkbox"/> Coefficient of determination $R^2$  | $0.75 / 0.50 / 0.25$            | Substantial / moderate / weak         |
| <input type="checkbox"/> Effect size $f^2$                   | $0.02 / 0.15 / 0.35$            | Small / medium / large                |
| <input type="checkbox"/> Predictive relevance $Q^2$          | $> 0$ via blindfolding          | Or use $PLSpredict$ (Shmueli et al.)  |
| <input type="checkbox"/> Model fit (SRMR)                    | $< 0.08$ ( $< 0.10$ acceptable) | Report $d_{ULS}$ , $d_G$ bootstrap CI |

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## STEP 4 REPORTING ESSENTIALS (methods section must-haves)

| METRIC                                               | THRESHOLD                  | IF IT FAILS                                                                  |
|------------------------------------------------------|----------------------------|------------------------------------------------------------------------------|
| <input type="checkbox"/> Software + version          | AnalyVa 4.2.2 / SmartPLS 4 | <i>Cite it (see <a href="https://analyva.com/cite">analyva.com/cite</a>)</i> |
| <input type="checkbox"/> Bootstrap settings          | 5,000 subsamples · BCa     | <i>State one- vs two-tailed a priori</i>                                     |
| <input type="checkbox"/> Random seed                 | Report for reproducibility | <i>AnalyVa exports the seed automatically</i>                                |
| <input type="checkbox"/> Sample size + justification | Inverse sqrt / G*Power     | <i>Power for smallest expected effect</i>                                    |
| <input type="checkbox"/> Missing data treatment      | Mean / EM / listwise       | <i>Report % missing per indicator</i>                                        |
| <input type="checkbox"/> Common method bias          | Harman's / marker variable | <i>Add procedural remedies to design</i>                                     |
| <input type="checkbox"/> Model specification         | Reflective vs formative    | <i>Justify each choice theoretically</i>                                     |

## STEP 5 ADVANCED ANALYSES (report if performed)

| METRIC                                                 | THRESHOLD                 | IF IT FAILS                            |
|--------------------------------------------------------|---------------------------|----------------------------------------|
| <input type="checkbox"/> Mediation (indirect effects)  | VAF / Zhao et al.         | <i>Bootstrap indirect effect CI</i>    |
| <input type="checkbox"/> Moderation (interaction)      | Two-stage or orthog.      | <i>Report simple slopes plot</i>       |
| <input type="checkbox"/> MGA (multi-group analysis)    | MICOM invariance first    | <i>Then permutation / Henseler MGA</i> |
| <input type="checkbox"/> IPMA (importance-performance) | Rescale on 0–100 scale    | <i>Importance vs performance chart</i> |
| <input type="checkbox"/> FIMIX / PLS-POS segmentation  | AIC3 / EN / BIC for k     | <i>Report profile of each segment</i>  |
| <input type="checkbox"/> Endogeneity (Gaussian copula) | Ramsey RESET / Park-Gupta | <i>Correct via Gaussian copula</i>     |

### HOW TO USE THIS CHECKLIST

Print it. Keep it beside your keyboard while you write. Tick each box as you write the corresponding sentence in the methods or results section. If a box stays empty, you owe your reviewer an explanation for why — or you owe your model an extra run.

Grounded in Hair et al. (2022), Henseler et al. (2015), and Sarstedt et al. (2020).

Compiled by Abdelouahd Bouzar, PhD

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